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APPOINTMENTS

Rice University
Assistant Professor, Department of Civil & Environmental Engineering — 2021—
Affiliated Faculty, Severe Storm Prediction, Education, and Evacuation from Disasters (SSPEED) Center — 2021—
Affiliated Faculty, Ken Kennedy Institute — 2023—

The Pennsylvania State University
Postdoctoral Scholar, Earth & Environmental Systems Institute — 2020

EDUCATION

Columbia University
Ph.D. in Earth & Environmental Engineering — 2020
M.S. in Earth & Environmental Engineering — 2016

Yale University
B.S. in Mechanical Engineering — 2015

AWARDS AND HONORS

NSF CAREER Award, National Science Foundation — 2026
Outstanding Reviewer Award, Earth's Future — 2023
Nickolas and Liliana Themelis Fellowship, Fu Foundation School of Engineering and Applied Science, Columbia University — 2018
Graduate Research Fellowship, Climate and Large-Scale Atmospheric Dynamics, National Science Foundation — 2017
Presidential Distinguished Fellowship, Fu Foundation School of Engineering and Applied Science, Columbia University — 2015
Distinction in Major, Department of Mechanical Engineering and Materials Science, Yale University — 2015
Legacy Award, New Haven Promise — 2015
Larry Coben '79 Fellowship, Yale University — 2014
Vance-Carter Travel Award, Yale University — 2013
Thomas C. Barry Travel Award, Yale University — 2012

ADVISEE AWARDS

Dongwook Kim: Karen and John Huff Graduate Fellowship in Civil and Environmental Engineering. — 2025
Yuchen Lu: H.W. Reeves Endowed Scholarship. — 2022

PUBLICATIONS

† **denotes Rice advisee publication**. Google Scholar *citations*: 841, *h-index*: 14, *i10-index*: 17.

IN PREP. / UNDER REVIEW

[1] Pollack, A., Benedict, J., Deb, M., **Doss-Gollin, J.**, Judi, D., Lehman, W., Lutz, N., Reesman, C., Sarazen, E., Son, Y., Srikrishnan, V., Sun, N., and Keller, K. : *Unrefined National Building Inventories Can Mislead Risk Assessments and Decisions.* . DOI: 10.2139/ssrn.5575271 — —

JOURNAL ARTICLES

[28] Baer, J. A., Sebastian, A., Grimley, L. E., **Doss-Gollin, J.**, Wright, D. B., Hussain, M. A., and Webber, M. K. : "Neglecting Spatiotemporal Rainfall Variability Misrepresents Flood Hazard and Risk". *npj Natural Hazards*. DOI: 10.1038/s44304-026-00208-5 2026

[27] Hancock, C. L., Dee, S. G., Haider, M. R., **Doss-Gollin, J.**, Lehner, F., Murphy, K., and Muñoz, S. E. : "Twenty-First-Century Hydrological Trends in the Mississippi River Basin Intensify the East–West Moisture Gradient". *Journal of Climate*. DOI: 10.1175/jcli-d-25-0340.1 2026

[26] Murphy, K., Dee, S., Hancock, C., Pitchon, E., **Doss-Gollin, J.**, Wallace, E., and Muñoz, S. : "The Changing Relationship between the Bermuda High, Great Plains Low-Level Jet, and Mississippi River Basin Hydroclimate from the Last Millennium to 2100". *Journal of Geophysical Research: Atmospheres*. DOI: 10.1029/2025JD044340 2026

[25] O'Donnell, M., Dee, S., **Doss-Gollin, J.**, and Muñoz, S. E. : "Hydrologic Whiplash in the Mississippi River Basin: Mechanisms and Projections". *Geophysical Research Letters*. DOI: 10.1029/2026GL122807 2026

[24] Pollack, A. B., Auermuller, L., Burleyson, C. D., Campbell, J., Condon, M., Cooper, C., Coronese, M., Dangendorf, S., **Doss-Gollin, J.**, Hegde, P., Helgeson, C., Kopp, R. E., Kwakkel, J., Lesk, C., Mankin, J., Nicholas, R. E., Rice, J., Roth, S., Srikrishnan, V., Scheeler, M., Tuana, N., Vernon, C., Zhao, M., and Keller, K. : "Unlocking the Benefits of Transparent and Reusable Science for Climate Risk Management". *Proceedings of the National Academy of Sciences*. DOI: 10.1073/pnas.2422157123 2026

[23] Haider, M. R., Dee, S., **Doss-Gollin, J.**, Dunne, K., and Muñoz, S. : "Impact of 21st Century Climate Change on Mississippi River Basin Discharge in CESM2 Large Ensemble Projections". *Global and Planetary Change*. DOI: 10.1016/j.gloplacha.2025.104742 2025

[22] Liu, C., Kowal, D. R., **Doss-Gollin, J.**, and Vannucci, M. : "Bayesian Functional Graphical Models with Change-Point Detection". *Computational Statistics and Data Analysis*. DOI: 10.1016/j.csda.2024.108122 2025

[21] Liu, Y., **Doss-Gollin, J.**, Dai, Q., Veeraraghavan, A., and Balakrishnan, G. : "Downscaling Extreme Precipitation with Wasserstein Regularized Diffusion". *IEEE Transactions on Geoscience and Remote Sensing*. DOI: 10.1109/TGRS.2025.3611872 2025

[20] **Lu, Y.**†, Seiyon Lee, B., and **Doss-Gollin, J.**: "Bayesian Spatiotemporal Nonstationary Model Quantifies Robust Increases in Daily Extreme Rainfall across the Western Gulf Coast". *Environmental Research: Climate*. DOI: 10.1088/2752-5295/adf56e 2025

[19] O'Donnell, M., Murphy, K., **Doss-Gollin, J.**, Dee, S., and Munoz, S. : "Evaluation of Hydroclimatic Biases in the Community Earth System Model (CESM1) within the Mississippi River Basin". *Hydrology and Earth System Sciences*. DOI: 10.5194/hess-29-4637-2025 2025

[18] Pollack, A., **Doss-Gollin, J.**, Srikrishnan, V., and Keller, K. : "UNSAFE: An UNcertain Structure and Fragility Ensemble Framework for Property-Level Flood Risk Estimation". *Journal of Open Source Software*. DOI: 10.21105/joss.07527 2025

[17] Kazadi, A., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : "FloodGNN-GRU: A Spatio-Temporal Graph Neural Network for Flood Prediction". *Environmental Data Science*. DOI: 10.1017/eds.2024.19 2024

[16] Murphy, K., Dee, S., **Doss-Gollin, J.**, Dunne, K. B. J., O'Donnell, M., and Muñoz, S. : "Competing Influences of Land Use and Greenhouse Gas Emissions on Mississippi River Basin Hydroclimate Simulated over the Last Millennium". *Paleoceanography and Paleoclimatology*. DOI: 10.1029/2024PA004902 2024

[15] Singh, D., Bekris, Y. S., Rogers, C. D. W., **Doss-Gollin, J.**, Coffel, E. D., and Kalashnikov, D. A. : "Enhanced Solar and Wind Potential during Widespread Temperature Extremes across the U.S. Interconnected Energy Grids". *Environmental Research Letters*. DOI: 10.1088/1748-9326/ad2e72 2024

[14] Amonkar, Y., **Doss-Gollin, J.**, Farnham, D. J., Modi, V., and Lall, U. : "Differential Effects of Climate Change on Average and Peak Demand for Heating and Cooling across the Contiguous USA". *Communications Earth & Environment*. DOI: 10.1038/s43247-023-01048-1 2023

[13] Amonkar, Y., **Doss-Gollin, J.**, and Lall, U. : "Compound Climate Risk: Diagnosing Clustered Regional Flooding at Inter-Annual and Longer Time Scales". *Hydrology*. DOI: 10.3390/hydrology10030067 2023

[12] **Doss-Gollin, J.**, Amonkar, Y., **Schmeltzer, K.**†, and Cohan, D. : "Improving the Representation of Climate Risks in Long-Term Electricity Systems Planning: A Critical Review". *Current Sustainable/Renewable Energy Reports*. DOI: 10.1007/s40518-023-00224-3 2023

[11] **Doss-Gollin, J.** and Keller, K. : "A Subjective Bayesian Framework for Synthesizing Deep Uncertainties in Climate Risk Management". *Earth's Future*. DOI: 10.1029/2022EF003044 — 2023

[10] Garcia, M., Juan, A., **Doss-Gollin, J.**, and Bedient, P. : "Leveraging Mesh Modularization to Lower the Computational Cost of Localized Updates to Regional 2D Hydrodynamic Model Outputs". *Engineering Applications of Computational Fluid Mechanics*. DOI: 10.1080/19942060.2023.2225584 — 2023

[9] Wutich, A., Thomson, P., Jepson, W., Stoler, J., Cooperman, A. D., **Doss-Gollin, J.**, Jantrania, A., Mayer, A., Nelson-Nuñez, J., Walker, W. S., and Westerhoff, P. : "MAD Water: Integrating Modular, Adaptive, and Decentralized Approaches for Water Security in the Climate Change Era". *WIREs Water*. DOI: 10.1002/wat2.1680 — 2023

[8] Zhou, X., Duenas-Osorio, L., **Doss-Gollin, J.**, Liu, L., Stadler, L., and Li, Q. : "Mesoscale Modeling of Distributed Water Systems Enables Policy Search". *Water Resources Research*. DOI: 10.1029/2022WR033758 — 2023

[7] **Doss-Gollin, J.**, Farnham, D. J., Lall, U., and Modi, V. : "How Unprecedented Was the February 2021 Texas Cold Snap?". *Environmental Research Letters*. DOI: 10.1088/1748-9326/ac0278 — 2021

[6] **Doss-Gollin, J.**, Farnham, D. J., Ho, M., and Lall, U. : "Adaptation over Fatalism: Leveraging High-Impact Climate Disasters to Boost Societal Resilience". *Journal of Water Resources Planning and Management*. DOI: 10.1061/(asce)wr.1943-5452.0001190 — 2020

[5] **Doss-Gollin, J.**, Farnham, D. J., Steinschneider, S., and Lall, U. : "Robust Adaptation to Multiscale Climate Variability". *Earth's Future*. DOI: 10.1029/2019ef001154 — 2019

[4] Rözer, V., Kreibich, H., Schröter, K., Müller, M., Sairam, N., **Doss-Gollin, J.**, Lall, U., and Merz, B. : "Probabilistic Models Significantly Reduce Uncertainty in Hurricane Harvey Pluvial Flood Loss Estimates". *Earth's Future*. DOI: 10.1029/2018ef001074 — 2019

[3] **Doss-Gollin, J.**, Muñoz, Á. G., Mason, S. J., and Pastén, M. : "Heavy Rainfall in Paraguay during the 2015/16 Austral Summer: Causes and Subseasonal-to-Seasonal Predictive Skill". *Journal of Climate*. DOI: 10.1175/JCLI-D-17-0805.1 — 2018

[2] Farnham, D. J., **Doss-Gollin, J.**, and Lall, U. : "Regional Extreme Precipitation Events: Robust Inference from Credibly Simulated GCM Variables". *Water Resources Research*. DOI: 10.1002/2017WR021318 — 2018

[1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : "Analytic Modeling of Rainwater Harvesting in the Brazilian Semiarid Northeast". *Journal of the American Water Resources Association*. DOI: 10.1111/1752-1688.12376 — 2016

PEER-REVIEWED CONFERENCE PAPERS

[4] Kazadi, A., **Doss-Gollin, J.**, and Silva, A. : "Pluvial Flood Emulation with Hydraulics-Informed Message Passing". *Forty-First International Conference on Machine Learning*. URL: https://openreview.net/forum?id=kIHIA6Lr0B — 2024

[3] Kazadi, A. N., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : "Flood Prediction with Graph Neural Networks". *Climate Change AI*. URL: https://www.climatechange.ai/papers/neurips2022/75 — 2022

[2] Araújo Júnior, L. M., de Souza Filho, F. d. A., da Silva Silveira, C., Aragão Dias, T., and **Doss-Gollin, J.**: "Análise Dos Eventos de Seca No Nordeste Setentrional Brasileiro Com Base No Índice de Precipitação Normalizada". *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: 10.13140/rg.2.1.4610.7685 — 2014

[1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : "Considerações Sobre a Sustentabilidade Hídrica de Cisternas Para Captação de Chuva No Semiárido Brasileiro". *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: 10.13140/rg.2.1.4086.4807 — 2014

SPONSORED RESEARCH

Amounts reflect Rice portion for collaborative grants and subawards; total amount for direct awards.

PI, National Science Foundation – Infrastructure Systems and People. *CAREER: Drain or Retain? Strategic Assessment of Integrated Flood Infrastructure Portfolios with Physics-Informed ML*. No. 2543303. ($599,581) — 2026–2031

Co-PI, National Science Foundation – Confronting Hazards, Impacts and Risks for a Resilient Planet (CHIRRP). *RAISE: Flood Resilience in Rural Texas Communities* PI: Avantika Gori, with Andrew Juan and Keri Stephens. No. 2440167. ($999,986) 2025–2028

PI, Consortium for Enhancing Resilience and Catastrophe Modeling (CERCAT). *A Nonstationary Joint Probability Method for Tropical Cyclone Hazard Assessment* with Avantika Gori. No. P12. ($75,000) 2025–2026

PI, Ken Kennedy Institute at Rice University. *Advancing AI for Climate Risk and Urban Resilience* with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. ($80,000) 2025–2026

Co-PI, NVIDIA. *Computing Infrastructure for AI-enhanced Climate Risk and Resilience at Rice* PI: Arlei Lopes da Silva, with Avantika Gori. in-kind 2025–2025

PI, Ken Kennedy Institute at Rice University. *Advancing AI for Climate Risk and Urban Resilience* with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. ($80,000) 2024–2025

Co-PI, National Science Foundation – Industry-University Cooperative Research Centers (IUCRC). *IUCRC Planning Grant Rice University: Center for Climate, Equity and Resilience in Catmodeling (CERCat)* PI: Jamie Padgett, with Katherine Ensor, Avantika Gori, Leonardo Dueñas Osorio, Brian Davison, David Casagrande, Maryam Rahnemoonfar, Paolo Bocchini, and Yi-Chen Yang. No. 2413382. ($20,000) 2024–2025

Co-PI, Rice University Sustainability Institute. *Workshop on Nature-Based Solutions for Resilient Coastal Cities* PI: Philip Bedient, with Avantika Gori. ($7,500) 2024–2025

PI, Texas Water Development Board. *Developing Future Rainfall Frequency Grids for the State of Texas* with John Nielsen-Gammon. (Full grant $192,828. Subaward $77,750) 2022–2026

Co-PI, National Science Foundation – Climate and Large-Scale Dynamics (CLD). *Collaborative Research: Evaluating the Past and Future of Mississippi River Hydroclimatology to Constrain Risk via Integrated Climate Modeling, Observations, and Reconstructions* PI: Sylvia Dee, with Samuel Muñoz. No. 2147781. ($472,024) 2022–2025

PI, National Science Foundation – Strengthening America's Infrastructure (SAI). *Collaborative Research: EAGER: Participatory Design for Water Quality Monitoring of Highly Decentralized Water Infrastructure Systems* with Alicia Cooperman, Alex Mayer, Shane Walker, and Manuela Muñoz Fuerte. No. 2120829. ($85,046) 2022–2025

Co-PI, 100,000 Strong in the Americas Innovation Fund. *IFCE-Rice-SENAI Program on Artificial Intelligence for Urban Sustainability and Resilience to Natural Disasters in the Americas* PI: Arlei Lopes da Silva. ($50,000) 2022–2023

Co-PI, Energy Foundation. *Synthesis of Texas Electricity Research from Rice University* PI: Daniel Cohan. ($24,928) 2022–2023

PI, Rice University – Sustainable Futures Fund (SFF). *Leveraging Earth System Observations at Multiple Scales to Improve Stormwater Management in Houston* with Sylvia Dee. ($50,000) 2022–2023

PRESENTATIONS

INVITED TALKS

"Statistical Modeling for Nonstationary Hydroclimate Hazards: Challenges and Opportunities". Statistics Colloquium, **Rice University**, Houston, TX. 2025

Panelist. Exploring Water Resilience to Emerging Challenges, **Society for Risk Analysis**, Remote Presentation. 2025

"Advancing Urban Flood Risk Management through Physics-informed, Data-Driven Hazard Assessment". Earth, Marine, and Environmental Science Seminar, **University of North Carolina**, Chapel Hill, NC. 2024

"Quantifying and Characterizing Uncertain Climate Hazards to Enable Adaptive Resilience". Atmospheric Sciences Seminar, **Texas A&M University**, College Station, TX. 2022

"Revisiting Our Design Criteria: What Hazards Should We Design for in a Changing Climate?". Hydrologic Sciences and Water Resources Engineering Seminar, **University of Colorado Boulder**, Remote Presentation. 2022

"Adapting Engineering Design Criteria to a Changing Climate: Insights from House Elevation ". Technical Webinar, **ASCE Central New Jersey Branch**, Remote Presentation. 2022

Panelist. Extreme Weather: How To Report on a World That's Warmer, Colder, Wetter, Drier and Weirder, **31st Annual Conference of the Society of Environmental Journalists**, Houston, TX. 2022

"Predictable and Preventable: Lessons in Climate Risk Management from the Texas Coast (and Beyond)". Climate Risk and Financial Institutions Seminar (Prof. Madison Condon), **Boston University School of Law**, Remote Presentation. 2022

"Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Blackouts". Outreach Event: Science is for Everyone, **American Meteorological Society**, Remote Presentation. 2022

"Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Blackouts". Compound Events Working Group, **Risk KAN (Knowledge Action Networks)**, Remote Presentation. 2021

Panelist. Tail events: Prediction, Planning, and Performance, **Harvard Electricity Policy Group**, Remote Presentation. 2021

"Towards Adaptive Resilience: Managing Flood Risks in a Changing World". Technical Webinar, **ASCE Central New Jersey Branch**, Remote Presentation. 2021

"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Center for Climate Risk Management CLIMA Seminar, **the Pennsylvania State University**, State College, PA. 2020

"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Department of Civil and Environmental Engineering Seminar, **Rice University**, Houston, TX. 2020

"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Complex Systems Simulation and Optimization Group, **National Renewable Energy Laboratory**, Golden, CO. 2020

"Drivers of Extreme Rainfall: Atmospheric Circulation Patterns and Regional Intense Rainfall in the Ohio River". European Flood Awareness System Group, **European Centre for Medium Range Weather Forecasting**, Reading, England. 2016

"Understanding the Physical Drivers of Extreme Rainfall for Flood Prediction". Oxford Water Network, **Oxford University**, Oxford, England. 2016

FIRST-AUTHOR CONFERENCE AND WORKSHOP PRESENTATIONS

"Industry-Academia Collaborations on Seasonal and Long-Term TC Hazard Assessment (CERCat)". **Catastrophe Risk Management Conference (CATRISK26)**, *Reinsurance Association of America*, Orlando, FL [*Panel*]. 2026

"H24A-07: TxRain: A Bayesian framework for integrating historical observations and model projections to develop nonstationary IDF curves". **American Geophysical Union Fall Meeting 2025**, New Orleans, LA [*Talk*]. 2025

"Robust Trends in Extreme Rainfall Probabilities in Texas". **2025 Texas Climate Conference**, *Rice University and Texas A&M University*, Houston, TX [*Talk*]. 2025

"Use-Inspired Tools for Climate Hazard Assessment". **Nature-Based Solutions for a Resilient Gulf Coast Workshop**, *Rice University*, Houston, TX [*Talk*]. 2025

"Advancing Urban Flood Hazard Characterization through Machine Learning: Challenges and Opportunities". **American Geophysical Union Fall Meeting 2024**, Washington, DC [*Talk*]. 2024

"Assessing and Managing Climate Risks to Electricity Systems in an Era of Climate Change and Energy Transition". **American Geophysical Union Fall Meeting 2024**, Washington, DC [*Talk*]. 2024

"Leveraging Machine Learning to Advance Urban Flood Hazard Assessment: Challenges and Opportunities". **Data-driven and physics-based machine learning methods for forecasting and knowledge discovery of surface hydrology**, *Conference on Computational Methods in Water Resources*, Tucson, AZ [*Talk*]. 2024

"Incorporating Temperature Projections into Energy Systems Planning". **Extreme Heat Workshop**, *Columbia University*, New York, NY [*Talk*]. 2024

"NH14B-07: Linking Robust Trends in Observations and Models to Develop Nonstationary Rainfall Frequency Grids for the State of Texas". **American Geophysical Union Fall Meeting 2023**, San Francisco, CA [*Talk*]. 2023

"A Bayesian Spatial Hierarchical Framework for Process-Informed Nonstationary Analysis of Precipitation Frequencies". **13th International Workshop on Statistical Hydrology**, *International Association of Hydrological Sciences*, Boston, MA [*Talk*]. 2023

"H35F-07: Near-term Predictability Lowers Long-Term Adaptation Costs". **American Geophysical Union Fall Meeting 2022**, Chicago, IL [*Talk*]. 2022

"Unprecedented Impacts Don't Require Unprecedented Weather". **Post-Harvey Climate & Flood Impacts on the Built Environment**, *Severe Storm Prediction, Education, & Evacuation from Disasters Center*, Houston, TX. 2022

"H25U-1265: Operationalizing Bayesian Model Checking for Robust Decision Making: Insights from House Elevation". **American Geophysical Union Fall Meeting 2021**, New Orleans, LA [*Poster*]. 2021

"A14H-03: How Unprecedented Was the February 2021 Texas Cold Snap?". **American Geophysical Union Fall Meeting 2021**, New Orleans, LA [*Talk*]. 2021

"Valuing Flexibility and Soft Instruments for Sequential Decision Problems". **2020 Annual Meeting**, *Society for Decision Making under Deep Uncertainty*, [*Remote Presentation*]. 2020

"H11G-07: Towards Adaptive Resilience: Managing Uncertainties and Exploiting Predictability across Timescales". **American Geophysical Union Fall Meeting 2019**, San Francisco, CA [*Talk*]. 2019

"Adaptive Resilience through Real Options and Deep Reinforecement Learning". **Doctoral Consortium on Computational Sustainability**, *Carnegie Mellon University*, Pittsburgh, PA [*Talk*]. 2019

"Evaluating Staged Investments in Critical Infrastructure for Climate Adaptation". **2019 Interdisciplinary Ph.D. Workshop in Sustainable Development**, *Columbia University*, New York, NY [*Talk*]. 2019

"H52F-05: Robust Adaptation to Cyclical Climate Risk". **American Geophysical Union Fall Meeting 2018**, Washington, DC [*Talk*]. 2018

"Robust Adaptation to Multi-Scale Climate Variability". **The Nexus of Climate Data, Insurance, and Adaptive Capacity**, Asheville, NC [*Poster*]. 2018

"A31H-0135: Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". **American Geophysical Union Fall Meeting 2017**, New Orleans, LA [*Talk*]. 2017

"H22B-02: Designing and Operating Infrastructure for Nonstationary Flood Risk Management". **American Geophysical Union Fall Meeting 2017**, New Orleans, LA [*Talk*]. 2017

"Extreme Rainfall in Paraguay during the 2015-16 Austral Summer: Causes and Predictive Skill". **North East Graduate Student Water Symposium**, *University of Massachusetts Amherst*, Amherst, MA [*Talk*]. 2017

"Regional Intense Precipitation: Inferences From GCM Atmospheric Circulation Fields". **Modeling Research in the Cloud**, *National Center for Atmospheric Research*, Boulder, CO [*Poster*]. 2017

"Statistical-Dynamical Analysis of Climate Projections for Flood Infrastructure Design". **Interdisciplinary Ph.D. Workshop in Sustainable Development 2017**, *Columbia University*, New York, NY [*Talk*]. 2017

"Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". **Workshop on Subseasonal to Seasonal Predictability of Extreme Weather and Climate**, *Columbia University*, New York, NY [*Poster*]. 2016

ADVISEE CONFERENCE AND WORKSHOP PRESENTATIONS

Dongwook Kim†: "Evaluating the Generalization of Sentinel-1 Flood Mapping across Diverse Environments". **Fall Meeting**, *Consortium for Enhancing Resilience and Catastrophe Modeling (CERCat)*, Bethlehem, PA [*Poster*]. 2026

Dongwook Kim†: "Comparative Analysis of Satellite-Based Flood Mapping: A Case Study of Hurricane Harvey". **39th Annual International Conference on Coastal Engineering (ICCE)**, *Coasts, Oceans, Ports & Rivers Institute (COPRI) - ASCE*, Galveston, TX [*Talk*]. 2026

Yuchen Lu†: "NH13D-0410: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". **American Geophysical Union Fall Meeting 2025**, New Orleans, LA [*Poster*]. 2025

Yuchen Lu†: "TxRAIN-Observational: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". **American Geophysical Union Fall Meeting 2024**, Washington, DC [*Talk*]. 2024

Yuchen Lu†: "Nonstationary Extreme Precipitation Probabilities in Texas". **Fall Meeting**, *Consortium for Enhancing Resilience and Catastrophe Modeling (CERCat)*, Bethlehem, PA [*Poster*]. 2024

Yuchen Lu†: "Spatially Varying and Duration Dependent Covariate Model: A Hierarchical Bayesian Framework for Multi-duration Extreme Precipitation Frequency Analysis in Texas". **World Environmental & Water Resources Congress 2024**, *Environmental & Water Resources Institute (EWRI) - ASCE*, Milwaukee, WI [*Talk*]. 2024

Yuchen Lu†: "H21T-1602: Spatially Varying Covariate Model: A Hierarchical Bayesian Framework for Precipitation Frequency Analysis in the Gulf Coast". **American Geophysical Union Fall Meeting 2023**, San Francisco, CA [*Talk*]. 2023

Yuchen Lu†: "H42E-1333: Nonstationary GEV with Hierarchical Spatial Pooling: A Spatiotemporal Bayesian Framework for Nonstationary Extreme Precipitation Frequency Analysis in the Gulf Coast". **American Geophysical Union Fall Meeting 2022**, Chicago, IL [*Talk*]. 2022

TEACHING AND ADVISING

COURSES TAUGHT

Rice CEVE 543: *Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes.* Fall 2026
Rice CEVE 421/521: *Climate Risk Management.* Spring 2026
Rice CEVE 543: *Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes.* Fall 2025
Rice CEVE 101: *Fundamentals of Civil and Environmental Engineering.* Fall 2024
Rice CEVE 421/521: *Climate Risk Management.* Spring 2024
Rice CEVE 543: *Data Science Methods for Climate Hazard Assessment.* Fall 2023
Rice CEVE 421/521: *Climate Risk Management.* Spring 2023
Rice CEVE 543: *Environmental Data Science.* Spring 2022
Rice CEVE 101: *Fundamentals of Civil and Environmental Engineering.* Fall 2021

CURRENT GRADUATE STUDENTS

Andres Calvo: **Ph.D.** in Civil and Environmental Engineering, Rice University. **Committee Member**.

True Furrh: **Ph.D.** in Civil and Environmental Engineering, Rice University. **Committee Member**. 2021—

Dongwook Kim: **Ph.D.** in Civil and Environmental Engineering, Rice University. **Primary Advisor**. 2024—

Kelsey Murphy: **Ph.D.** in Earth, Environmental, and Planetary Sciences, Rice University. **Committee Member**. 2021—

Jiayue Yin: **Ph.D.** in Earth, Environmental, and Planetary Sciences, Rice University. **Committee Member**. 2023—

PAST GRADUATE STUDENTS

Primary Advisor

Yuchen Lu: **Ph.D.** in Civil and Environmental Engineering, Rice University. Thesis: *Probabilistic Analysis of Nonstationary Hydroclimate Extremes.* 2026

Katlyn Schmeltzer: **MCEE** in Civil and Environmental Engineering, Rice University. Thesis: *Pre-Trained Long Short-Term Memory Network Performance for Streamflow Prediction in the Brazos River Basin.* 2025

Committee Member

Valeriia Sobolevskaia: **Ph.D.** in Earth, Environmental, and Planetary Sciences, Rice University. Thesis: *The seismic signatures of the hydrological cycle as observed by distributed acoustic sensing.* 2026

Karan Jakhar: **Ph.D.** in Mechanical Engineering, Rice University. Thesis: *Equation Discovery and Deep Learning for Geophysical Turbulence.* 2025

Kyle Ostlind: **M.S.** in Civil and Environmental Engineering, Rice University. Thesis: *Evaluating Runoff Response to Nature-Based Solutions under Varying Development Scenarios in Upper Cypress Creek near Houston, Texas.* 2025

John A. Baer: **M.S.** in Earth, Marine and Environmental Sciences, University of North Carolina at Chapel Hill. Thesis: *Quantifying Precipitation-Induced Uncertainty in Flood Hazard Assessment in a Coastal Urban Area.* 2024

Kendall Capshaw: **Ph.D.** in Civil and Environmental Engineering, Rice University. Thesis: *Modeling Coastal Petrochemical Infrastructure Risk, Resilience, and Cascading Community Consequences.* 2024

Xinyue Luo: **Ph.D.** in Earth, Environmental, and Planetary Sciences, Rice University. Thesis: *Characterizing the El Niño-Southern Oscillation and Its North American Teleconnections over the Last Millennium.* 2024

Anibal Tafur Gutierrez: **Ph.D.** in Civil and Environmental Engineering, Rice University. Thesis: *Methods and Tools for Risk-Informed Resilience Enhancement of Coastal Intermodal Freight Networks.* 2024

Matthew Garcia: **Ph.D.** in Civil and Environmental Engineering, Rice University. Thesis: *Novel Urban Floodplain Modeling Methods for Applications in Coupling Surrogate Machine Learning Methods.* 2023

Mia Peeples: **M.S.** in Civil and Environmental Engineering, Rice University. Thesis: *Modeling Flood Reduction of Nature-Based Channel Modifications in Houston, TX.* 2023

Xiangnan Zhou: **Ph.D.** in Civil and Environmental Engineering, Rice University. Thesis: *Resilience Planning for Water Distribution Systems.* 2023

Raychel Bahnick: **M.S.** in Civil and Environmental Engineering, Rice University. Thesis: *Assessing Land Use Change and Subsidence Impact on Inland Flooding*. 2022

Alyssa Graham: **M.S.** in Civil and Environmental Engineering, Rice University. Thesis: *Water Supply Vulnerability Testing and Robust Planning Analysis with Exploratory Modeling under Deep Uncertainty*. 2022

Elizabeth Hoffmann: **M.S.** in Civil and Environmental Engineering, Rice University. Thesis: *Mapping Dynamic Watershed Response Under Increasing Development Using HEC-RAS 2D: A Case Study of the Big Creek Watershed in Fort Bend County*. 2022

Chunshan Liu: **Ph.D.** in Statistics, Rice University. Thesis: *Bayesian Graphical Models for Multivariate Time Series.* 2022

Xiaoyu (Toby) Li: **M.S.** in Civil and Environmental Engineering, Rice University. Thesis: *Evaluating the Effects of Project Brays Mitigation Using Unsteady HEC-RAS Hydraulic Modeling: Application to Meyerland in Houston, TX.* 2021

UNDERGRADUATE RESEARCHERS

Year indicates graduation year.

Zain Rahman: *B.S. in Computer Science*, Rice University. 2027
Kyle Olcott: *B.S. in Civil and Environmental Engineering*, Rice University. 2025
Sophia Prieto: *B.S. in Statistics*, Rice University. 2023
John Cook: *B.S. in Civil and Environmental Engineering*, Rice University. 2022

TEAMS ADVISED

Optimal Policy for Decentralized Wastewater Systems (DWS) while Relaxing Certainty: Rice Computational Mathematics and Operations Research (CMOR) Senior Project. 2024-2025

Flood Sight | Advancing Real-Time Flood Predictions for Situational Awareness: Rice Data to Knowledge (D2K) Lab. Spring 2025

Predicting Streamflow for Hill Country Gauges Using a Transformer-based Model: Rice Data to Knowledge (D2K) Lab. Fall 2025

SERVICE ACTIVITIES

DEPARTMENTAL SERVICE

Member, Graduate Studies Committee. 2024—
Member, Diversity, Equity, and Inclusion Committee. 2024
Member, Faculty Search Committee. 2022–2023
Member, Seminar Committee. 2022–2023

UNIVERSITY SERVICE

Member, Research Council. Ken Kennedy Institute. 2024—
Faculty Associate. Duncan College. 2023—
External Search Committee Member. Department of Earth, Environmental, and Planetary Sciences. 2022–2023

PROFESSIONAL SERVICE

Working Group Member, Digital Twins - Beyond Physical Systems. Digital Earths Lighthouse Activity. 2026–2030

Committee Member, Water and Society Newsletter Committee. American Geophysical Union. 2024—

Committee Member, Environmental and Water Resources Systems (EWRS) Committee. American Society of Civil Engineers (ASCE) Environmental & Water Resources Institute (EWRI). 2021—

PEER REVIEW

Journals: AGU Advances; Climate Risk Management; Climatic Change; Communications Earth and Environment; Earth's Future; Energy Technology; Engineering Applications of Computational Fluid Mechanics; Environmental Data Science; Environmental Research Letters; Environmental Research: Climate; Geophysical Research Letters; Hydrology and Earth System Sciences; IEEE Transactions on Geoscience and Remote Sensing; Joule; Journal of Applied Meteorology and Climatology; Journal of Catastrophe Risk and Resilience; Journal of Hydrology; Journal of Water Resources Planning and Management; Machine Learning: Earth; Natural Hazards and Earth System Sciences; npj Natural Hazards; Oxford Development Studies; Water Resources Research; Water Security; Weather, Climate, and Society.

Funding Agencies: Department of Energy (BER); Dutch Research Council (NWO); National Science Foundation.

Other: Electric Power Research Institute (EPRI); Texas Water Development Board (TWDB).

SESSIONS CONVENED

Co-Convener. *NH43C - Advances in Urban Flood Risk Assessment and Adaptation IV Poster*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Convener. *NH43D - Advances in Urban Flood Risk Assessment and Adaptation V Poster*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Convener. *NH41E - Interdisciplinary Advances in Catastrophe Modeling and Disaster Resilience: Bridging Science, Policy, and Practice I Poster*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Convener. *NH43B - Interdisciplinary Advances in Catastrophe Modeling and Disaster Resilience: Bridging Science, Policy, and Practice II Oral*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Convener. *NH31A - Advances in Urban Flood Risk Assessment and Adaptation I Oral*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Convener. *NH32A - Advances in Urban Flood Risk Assessment and Adaptation II Oral*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Convener. *NH33A - Advances in Urban Flood Risk Assessment and Adaptation III Oral*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2025

Co-Organizer. *Nature-Based Solutions for a Resilient Gulf Coast*. **Rice University**, Houston, TX. 2025

Primary Convener. *H31G - Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security I Oral*. **American Geophysical Union Fall Meeting**, San Francisco, CA. 2023

Primary Convener. *H33P - Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security II Poster*. **American Geophysical Union Fall Meeting**, San Francisco, CA. 2023

Convener. *NH41C - Hybrid Modeling and Digital Twin Systems for Flood Hazard Prediction and Risk Assessment at Different Spatial Scales*. **American Geophysical Union Fall Meeting**, Washington, DC. 2023

Chair. *H44G - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making I Oral*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2021

Chair. *H45V - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making III Poster*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2021

Chair. *H45V - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making Poster Summary Session*. **American Geophysical Union Fall Meeting**, New Orleans, LA. 2021

Primary Convener. *NH53 – Emerging Needs and Approaches for Climate Services: Understanding and Developing Innovative Approaches to User-Oriented Climate Services*. **American Geophysical Union Fall Meeting**, San Francisco, CA. 2019

Student Organizer. *Earth and Environmental Engineering Student Research Symposium*. **Columbia University**, New York, NY. 2018

Student Organizer. *Earth and Environmental Engineering Student Research Symposium*. **Columbia University**, New York, NY. 2017

MEDIA HIGHLIGHTS

“FEMA elabora nuevos mapas de riesgo de inundación ¿Afectará esto tu propiedad?”. **Telemundo Houston** [*recorded video interview*]. 2026

“The Climate-Disaster Scores That Could Make or Break Your Home Sale”. **The Wall Street Journal** — Jean Eaglesham and Nicole Friedman [*print*]. 2026

“Historic Texas Flooding”. **KEYE-AUS (CBS)** [*recorded video interview*]. 2025

“Houston’s Morning Show”. **Fox26 Houston** [*live video interview*]. 2025

“Breaking down the force of water in the Texas floods”. **AP News** — Michael Phillis [*print*]. 2025

“Here are some things you can do to be better prepared for major flooding”. **Associated Press** — Caleigh Wells [*print*]. 2025

“ABC13-Luke Jones speaks with James Doss-Gollin, an assistant professor of Civil and Environmental Engineering at Rice University, about the key differences between flooding in Houston and central Texas.”. **ABC13** — Luke Jones [*recorded video interview*]. 2025

“After deadly flooding in Central Texas, state lawmakers look to prevent similar tragedies”. **KVUE** — Daniel Perreault [*recorded video interview*]. 2025

“New Flood Warning System Greenlit Shortly Before Deadly Texas Disaster”. **The Wall Street Journal** — Joseph De Avila [*print*]. 2025

“Questions arise on how emergency warning systems work after Central Texas flood”. **ABC13** — Tom Abrahams [*print*]. 2025

“Bajo la Amenaza del Golfo (Under the Threat of the Gulf)”. **Telemundo Houston** [*recorded video interview*]. 2025

“Will Texas become too hot for humans?”. **BBC Future** — Sarah Griffiths [*print*]. 2023

“Climate change has sent temperatures soaring in Texas”. **The Texas Tribune** — Erin Douglas, Yuriko Schumacher and Alex Ford [*print*]. 2023

“Texas could have foreseen 2021 cold-wave disaster, new study concludes”. **Texas Climate News** — Bob Henson [*print*]. 2022

“Opinion: The risks of climate change are great - so are the rewards of solving it”. **Houston Chronicle** — Andrew Dessler, James Doss-Gollin, and Katherine Hayhoe [*print*]. 2021

“The False Comfort of Higher Seawalls”. **The New Republic** — Paola Rosa-Aquino [*print*]. 2019

“New Study Shows Promise for Long-Term Weather Forecasts in South America”. **State of the Planet** — Elisabeth Gawthrop [*print*]. 2018

LANGUAGE SKILLS

English: Native
Spanish: Fluent
Portuguese: Proficient

TRAINING AND EXPERIENCE

Strategic Program to Accelerate Researchers in Computing (SPARC) Participant — *Natural Hazards Engineering Research Infrastructure (NHERI) DesignSafe-CI.* 2025

Panel Fellow — *NSF CMMI’s Game Changer Academies for Advancing Research Innovation.* 2021

Visiting Graduate Researcher — *Lamontagne Research Group, Department of Civil and Environmental Engineering, Tufts University, Medford, MA.* 2019–2020

Summer School Participant — *Fluid Dynamics of Sustainability and the Environment, Cambridge University, Cambridge, England.* 2016

Education Policy Intern — *Elm City Communities / New Haven Housing Authority, New Haven, CT.* 2015

President (2014), Design Lead (2013), Member (2012, 2015) — *Engineers Without Borders, Yale Student Chapter, New Haven, CT.* 2012–2015

Undergraduate Research Assistant — *Lab of Jaehong Kim, Department of Chemical and Environmental Engineering, Yale University, New Haven, CT.* 2014–2015

Visiting Undergraduate Researcher — *Water and Climate Risk Lab, Department of Hydraulic and Environmental Engineering, Universidade Federal do Ceará, Fortaleza, Brazil.* 2014

Mechanical Design Intern — *Slingshot Team, DEKA Research \& Development, Manchester, NH.* 2013

Undergraduate Research Assistant — *Lab of Jan Schroers, Department of Mechanical Engineering and Materials Science, Yale University, New Haven, CT.* 2012

Ikatú Agua Intern — *Fundación Paraguaya, Asunción, Paraguay.* 2012